

One-Page Summary Report – Online Retail Dataset

Introduction

The Online Retail dataset contains **541,909 transaction records and 8 columns** from a UK-based online retail business. The dataset provides information about invoices, products, quantities purchased, unit prices, transaction dates, customer IDs, and customer countries. The analysis focused on cleaning the data, validating its quality, and exploring sales trends, product demand, and customer distribution

Cleaning Challenges Encountered

The main cleaning challenges included **1,454 missing values in Description** and **135,080 missing values in CustomerID**. These were handled by replacing missing descriptions with "Unknown" and missing customer IDs with 0. The dataset also contained **5,268 duplicate rows**, which were removed. The `InvoiceDate` column was standardized to datetime format, while text formatting, column names, and data types were also standardized. Data validation identified **10,587 negative quantities**, **2 negative unit prices**, and **2,510 zero-price transactions**, which were treated as potential anomalies requiring further investigation.

Key EDA Findings

The exploratory analysis showed that the **United Kingdom is the dominant market**, with **3,951 customers**, significantly higher than other countries. Monthly sales fluctuated throughout the year, reaching a peak of approximately **1.45 million in November 2011**, followed by a sharp decline in December. Product analysis showed that **WORLD WAR 2 GLIDERS ASSTD DESIGNS** was the best-selling product, with **53,751 units sold**, followed by **JUMBO BAG RED RETROSPOT** with **47,260 units**.

Top Insights

- The **United Kingdom is the primary customer market**, with a significantly larger customer base than other countries.
- Sales activity **increased significantly toward the end of 2011**, with November recording the highest monthly sales.
- A small number of products showed **very high demand**, with the top-selling product exceeding 53,000 units sold.
- The dataset contains several **potential anomalies**, including negative quantities, negative prices, and zero-price transactions.
- The analysis indicates that the business could focus on its **high-performing products and core UK market** while exploring opportunities for international customer growth